

**To:** Dr. Anita Shah (Primary Care)

**From:** Hematology/Oncology Service

**Date:** 22.11.2023

**Patient:** Billy Peterson (born 11.11.52), MRN AML052

**Primary diagnosis:** Acute Myeloid Leukemia, **ELN 2022**

**Adverse, t(3q26.2;v)** consistent with MECOM(EVI1) activation

**First-line therapy:** Decitabine initiated **16.03.2023**

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## Notification of death

We are saddened to report that **Mr. Billy Peterson died on 16.11.2023** on our hematology service. The immediate cause of death was **septic shock with multifocal pneumonia** in the setting of **refractory AML** and profound treatment-related neutropenia. His code status was **DNR/DNI**, determined after family discussions reflecting his wishes.

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## Diagnostic background

- **Initial presentation (08.03.2023):** Fatigue, weight loss (~5 kg), gingival bleeding. Labs: **WBC  $10.9 \times 10^9/L$**  (blasts 18%), **Hgb 8.1 g/dL**, **Plt  $42 \times 10^9/L$** , LDH 520 U/L.
  - **Bone marrow:** 60% myeloblasts; immunophenotype CD34<sup>+</sup>/CD117<sup>+</sup>/HLA-DR variable, aberrant CD7.
  - **Cytogenetics: t(3q26.2;v);** FISH confirmed **MECOM** rearrangement. FLT3/NPM1 negative.
  - **Comorbidities:** COPD (ex-smoker, 40 pack-years), hypertension, CKD stage 3a, ischemic heart disease with prior stent (2014). Baseline echo LVEF 55%.
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## Treatment course

1. **Hypomethylating therapy** with **decitabine 20 mg/m<sup>2</sup> D1–5 q28d**, started **16.03.2023**. Early cycles complicated by prolonged cytopenias; one episode of febrile neutropenia managed with piperacillin-tazobactam. No TLS.
2. **Interim response:** After two cycles (May 2023), marrow showed **persistent disease** with ~25% blasts; we continued a third cycle given partial peripheral count improvement.

3. **Progression:** By **mid-October 2023**, circulating blasts reappeared (10–15%); marrow confirmed **refractory/relapsed AML** with 38% blasts. Options discussed included **venetoclax + azacitidine**, clinical trial, or best supportive care. Performance status had declined (ECOG 3) with frequent COPD exacerbations and weight loss; the patient prioritized quality of life and declined hospital-intensive salvage. We initiated **palliative-intent venetoclax monotherapy** at reduced dose with antimicrobial prophylaxis, but treatment was limited by infections and count suppression.
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### **Terminal admission (09.11.2023 → 16.11.2023)**

**Presentation:** Fever, productive cough, and dyspnea. Vitals: T 38.8 °C, BP 92/56 (MAP 63), HR 118, RR 28, SpO<sub>2</sub> 88% RA.

**Key labs:** WBC  $0.7 \times 10^9/L$  (ANC 0.03), Hgb 7.0 g/dL, Plt  $9 \times 10^9/L$ ; creatinine 1.9 mg/dL; AST/ALT 44/52; lactate 3.8 mmol/L; procalcitonin 9.8 ng/mL.

**Imaging:** CT chest — bilateral lower-lobe consolidations with ground-glass changes; moderate emphysema.

**Microbiology:** Blood cultures later positive for **ESBL E. coli**; sputum grew **Pseudomonas aeruginosa** susceptible to ceftolozane/tazobactam.

**Management:** Broad-spectrum IV antibiotics (meropenem → ceftolozane/tazobactam per sensitivities) + azithromycin, IV micafungin empirically, high-flow nasal oxygen, platelet and RBC transfusions, stress-dose steroids given COPD exacerbation. Despite escalation, he developed vasopressor-dependent shock and refractory hypoxemia. After a family meeting, he chose comfort-focused measures. He **passed away at 19:12 on 16.11.2023** with his daughter present.

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### **Medications near end of life**

Ceftolozane/tazobactam, micafungin, bronchodilators (nebulized), PRN hydromorphone for dyspnea, lorazepam PRN anxiety, acetaminophen, pantoprazole, bowel regimen. Chronic antihypertensives and statin were held.

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### **Aftercare**

Next of kin **Emily Peterson** (daughter) received bereavement resources. Autopsy declined. Death certificate completed; copies forwarded to your office.

Please accept our condolences and contact us if additional documentation is needed.

Sincerely,

**[Name], MD** — Consultant Hematologist